Requirements

3.1 Functional Requirements

Req#	Requirements	Description	Comments	Rate	Reviewed/Approved
FR_01	Production Batch Management	Logs and manages production batches including machine, shift, operator, and output details.	Ensures full traceability of production lifecycle.	High	Mei Dallashi Elgi Dhame Marisa Bajo
FR_02	Inventory Management	Tracks raw materials and finished goods stock levels in real time.	Prevents shortages and overstocking.	High	Mei Dallashi Elgi Dhame Marisa Bajo
FR_03	Sales Order Management	Creates and tracks sales orders from clients with full lifecycle status.	Ensures accurate and timely order fulfillment.	High	Mei Dallashi Elgi Dhame Marisa Bajo
FR_04	Delivery Management System	Generates delivery manifests and allows drivers to confirm deliveries.	Improves logistics clarity and accountability.	High	Mei Dallashi Elgi Dhame Marisa Bajo
FR_05	Purchase Order Management	Creates and tracks purchase orders for suppliers and logs received materials.	Supports continuous supply chain operations.	High	Mei Dallashi Elgi Dhame Marisa Bajo
FR_06	Workforce Shift Scheduling	Assigns employees to shifts, roles, and production lines.	Prevents scheduling conflicts and gaps.	Mediu m	Mei Dallashi Elgi Dhame Marisa Bajo
FR_07	Packing Queue Management	Displays and prioritizes orders that need packing.	Reduces dispatch errors and improves workflow efficiency.	Mediu m	Mei Dallashi Elgi Dhame Marisa Bajo
FR_08	Reporting System	Generates operational reports such as yield, cost per unit, and delivery performance.	Enables data-driven decision-making.	Mediu m	Mei Dallashi Elgi Dhame Marisa Bajo
FR_09	Low Stock Alert System	Automatically notifies when stock falls below minimum levels.	Supports proactive inventory management.	Low	Mei Dallashi Elgi Dhame Marisa Bajo
FR_10	Machine Output History Tracking	Stores and displays historical production data per machine.	Helps detect inefficiencies and maintenance needs.	Low	Mei Dallashi Elgi Dhame Marisa Bajo

Req#	Requirements	Description	Comments	Rate	Reviewed/Approved
FR_11	Supplier Management	Maintains supplier records and links them to purchase orders and materials.	Ensures traceability of supply chain sources.	High	Mei Dallashi Elgi Dhame Marisa Bajo
FR_12	Client Management	Stores client details and links them to sales orders and deliveries.	Centralizes customer data for operations.	High	
FR_13	Product and Material Management	Manages definitions of raw materials and finished products.	Ensures consistency across production and inventory.	High	Mei Dallashi Elgi Dhame Marisa Bajo
FR_14	Audit and Traceability System	Maintains logs for all operations including production, inventory, and orders.	Guarantees accountability and data integrity.	High	Mei Dallashi Elgi Dhame Marisa Bajo

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3.2 Non-Functional Requirements

3.2.1 Product Requirements

3.2.1.1 Security Requirements

Encryption of sensitive operational and employee data at rest and in transit.

Role-based access control (RBAC) for operators, managers, and drivers.

Secure authentication for all system users.

Logging and monitoring of all system activities.

Regular backups to prevent data loss.

3.2.1.2 Performance Requirements

System response time should be under 3 seconds for standard operations.

Real-time updates for inventory and production data.

Support concurrent users across departments without degradation.

3.2.1.3 Reliability Requirements

Automated daily backups of operational data.

Failover mechanisms for database availability.

Minimal downtime with recovery mechanisms in case of system failure.

3.2.2 Organizational Requirements

1. Training and Documentation

User manuals for operators, managers, and logistics staff.

Training sessions for employees on system usage.

Integrated help/support system.

2. Change Management

Procedures for updates and system improvements.

Version control and rollback mechanisms.

3.2.2.1 Environmental Requirements

1. Technologies Used

Backend: Java / Node.js

Frontend: HTML, CSS, JavaScript

Database: PostgreSQL / MySQL

2. Infrastructure and Hardware

Servers capable of handling real-time operations

Workstations for factory staff

Barcode scanners (optional for inventory)

3. Software Dependencies

Database management system

Server environment (Linux/Windows Server)

4. Network Requirements

Stable internal network for factory operations

Secure connections for remote access

5. Data Storage and Management

Storage for production, inventory, and logistics data

Backup and retention policies

6. Environmental Security

Physical protection of servers and IT infrastructure

Monitoring access to systems

7. Integration with External Systems

Potential integration with accounting or ERP systems

API support for third-party logistics

8. Scalability Testing Environment

Testing system performance under increased production load

9. Usability Testing Environment

Testing with factory staff for usability improvements

10. Training Facilities

On-site or digital training resources

3.2.3 External Requirements

1. Integration

Integration with supplier systems (optional)

Integration with logistics tracking systems

2. Regulatory Compliance

Compliance with business and labor regulations

Data protection compliance for employee data

3.2.3.1 Ethical Requirements

1. Fair Workforce Management

Transparent shift scheduling

Equal workload distribution

2. Responsible Data Use

Ethical handling of employee and operational data

3.2.3.2 Legislative Requirements

1. Data Protection and Privacy

Compliance with data protection laws (e.g., GDPR principles)

2. Labor Regulations

Compliance with workforce scheduling and working hours laws

3. Financial and Trade Regulations

Compliance with invoicing, sales, and procurement laws

3.3 Domain Requirements

1. Production Management

Track production batches linked to machines, shifts, and operators

Record output quantities and production status

2. Inventory and Supply Chain

Real-time stock tracking

Supplier-linked procurement system

3. Order and Delivery Management

Full lifecycle tracking of sales orders

Delivery confirmation and issue logging

4. Workforce Management

Shift scheduling with conflict detection

Role-based task assignment

5. Reporting and Analytics

Automated generation of operational KPIs

Filtering and exporting reports

6. Audit and Traceability

Full traceability of all operations

Logs for compliance and analysis

7. System Scalability

Ability to support increased production lines and users

8. Workflow Automation

Automated alerts, notifications, and status updates